

Limitations of Analytics over Anonymously Submitted Data Items

The most immediate limitation of using self-submitted data from open source websites comes from the way data is provided – through self-submission. Applicants choose to post their results, and the data from Grad Café is not truly representative of the total number of applicants, simply because not all applicants choose to submit their data on the website. Moreover, there may be a bias in the applicant data available. Student who are more proud of their acceptance, and conversely, students who are incredibly frustrated with either their rejection or waitlist or interview process, may be more likely to share their results on Grad Café. This bias would introduce inaccuracies in averages of GPAs and GRE scores because students who are proud of their score tend to be more likely to share them than students who did not perform as well academically.

Using self-submitted data, there is another limitation in the quality and reliability of provided data. Submissions to Grad Café are anonymous, thus there exists no method of verification of quantitative metrics. Students can submit wildly inaccurate metrics of GPA and GRE scores - even fabricate data - and Grad Café will accept the entry. Additionally, there is no ID associated with a student, so there are no protections in place to prevent duplications of entries. Another problem is that students can freely enter the university and program name as text, which gives room for spelling errors, thereby making data analysis trickier.

Despite the limitations of these open-source datasets, they can provide some useful insight. For example, question 11 in my assignment compared the GPAs of accepted and rejected PhD applicants. The result showed that there was not a large difference between the GPAs, indicating that GPA was not a differentiating factor between the pool of accepted and rejected PhD applicants.